

From the labs

Gone are the days when sensors were simple technology – these are the sensors of the future.

Space age

A Superluminous Supernova a.k.a Hypernova is a high energy star explosion that looks stunning as well as scary.

SOLAR ENERGY

IS THE TIME RIPE FOR THE DREAM NOW?



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Tesla's project of powering an entire American island using solar power, freeing the citizens from depending on diesel energy has revived the excitement surrounding solar power. Is it a one-off or has the time truly come for solar energy?

It's no secret that the sun is a great source of energy. After all, it is what powers all the living things on this planet! In fact, the sun showers on the earth enough energy to take care of the world's power requirements multiple times over.

On the plus side, it doesn't produce carbon dioxide emissions. So, you wouldn't have to worry about adverse environmental effects on that front. Another advantage is that the sun is practically inexhaustible. Also, the sun

doesn't charge any money for making use of its energy!

So, the question is how the people on earth can turn the sunlight into electricity that we can use?

How to harness energy from the sun

To do this, old-school solar technology makes use of large crystals that are made of silicon. These crystals generate electrical current when struck with light. The reason why silicon does this



A red boiling pot of energy

is because the electrons in the crystal move when they are exposed to light. (In many materials, the electrons would just jiggle around when struck by light—the reason why heat is generated.)

Inside a crystal, the bonds that exist between the silicon atoms are made of electrons which are shared among the atoms in the crystal. When the light is absorbed, sometimes an electron in a bond gets excited up to a high energy level and therefore is able to move more freely than when it was bound. This means that the electron can now freely move around in the crystal, resulting in a current in the crystal.

Silicon has the advantage of turning a decent amount of the light energy into electrical energy.

The downside to using silicon

The downside is that it's quite expensive since big crystals aren't easy to grow.